Chapter 13: Multi-Channel Sequence Analysis

1. Introduction

Multi-channel sequence analysis extends simple sequences to include multiple concurrent event streams. Students interact with courses through different channels (forums, quizzes, assignments). Understanding cross-channel patterns reveals how different learning modalities interact.

2. Dataset & Channels

Event sequences from 30 students across 3 channels (forum, quiz, assignment). Total events: 3000.

3. Methods

Events were extracted and timestamped. Cross-channel transition probabilities computed by examining consecutive events across all channels. Temporal patterns visualized.

4. Results

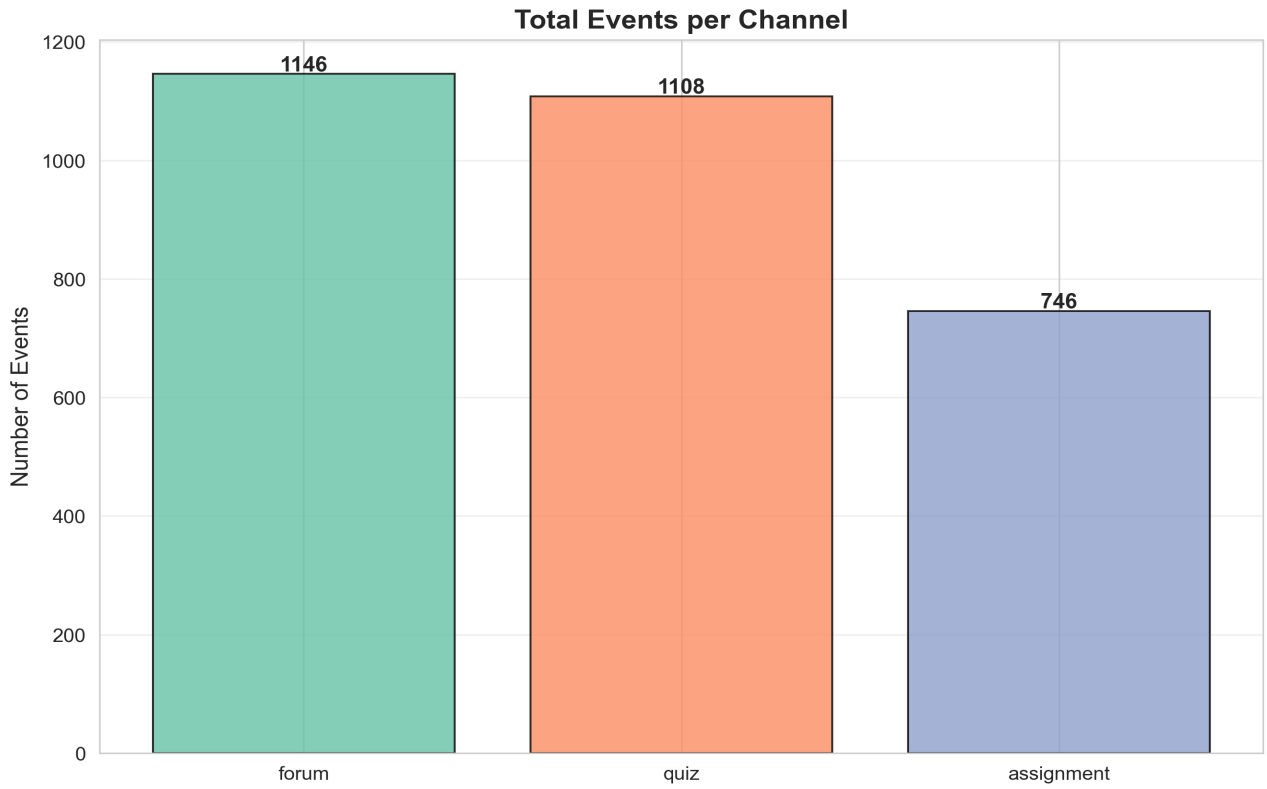

Fig 1: Event distribution across channels.

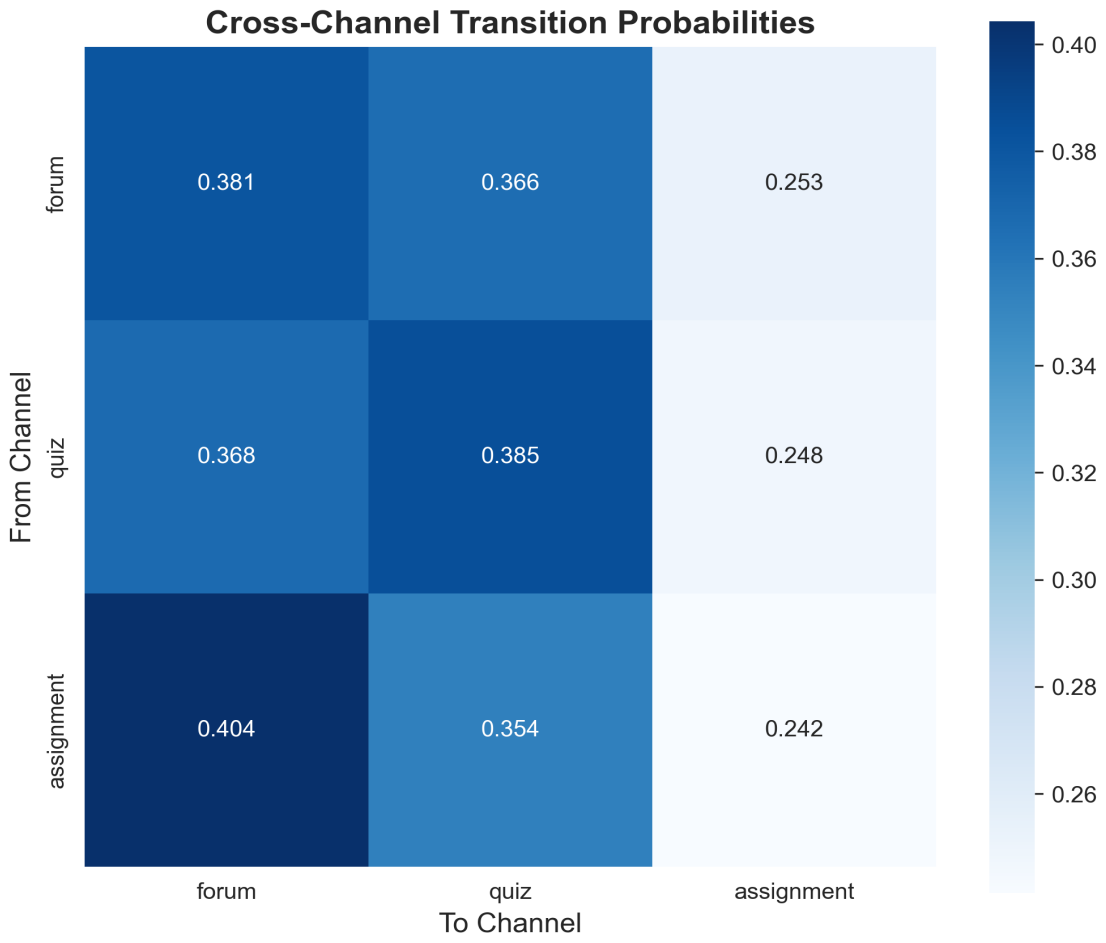

Fig 2: Cross-channel transition patterns.

5. Interpretation

Cross-channel analysis reveals students' preferred navigation patterns and learning pathways. High transition probabilities between specific channels indicate integrated learning activities.

6. Limitations

Analysis assumes channel types are independent, though students may employ strategic navigation. Time gaps between channels not explicitly modeled.

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